

Part 1: Data Preparation & Cleaning (1.5 hrs)

1. Consolidate Sales Data

- o Combine Sales_Q1, Sales_Q2, Sales_Q3 into one Sales_All sheet.

The screenshot shows the Power Query Editor interface. The main area displays a table with 7 columns: OrderID, Date, Region, CustomerID, Product, and Quantity. The table contains 20 rows of data. The 'Query Settings' pane on the right shows the 'Name' as 'Append1' and the 'Applied Steps' as 'Source' and 'Appended Query'.

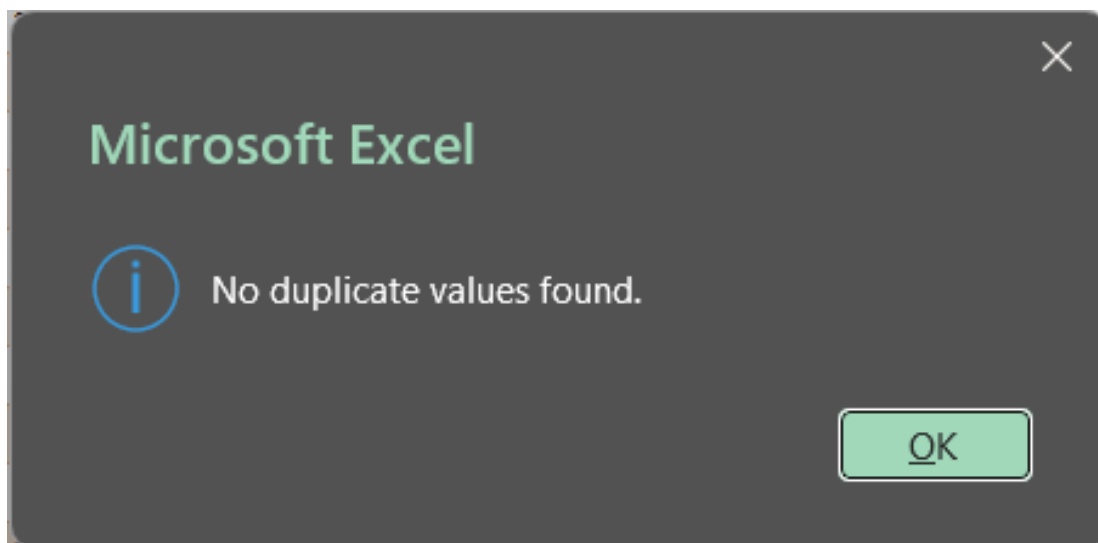
OrderID	Date	Region	CustomerID	Product	Quantity
Q1-001	45708,02709	South	1018	Tablet	
Q1-002	45712,23679	South	1004	Headphones	
Q1-003	45736,26829	West	1005	Monitor	
Q1-004	45703,5712	East	1018	Smartphone	
Q1-005	45663,68591	North	1012	Laptop	
Q1-006	45716,22677	East	1013	Laptop	
Q1-007	45735,11642	South	1017	Tablet	
Q1-008	45707,39413	South	1014	Tablet	
Q1-009	45671,49721	South	1020	Printer	
Q1-010	45743,20222	West	1010	Headphones	
Q1-011	45684,92907	West	1019	Laptop	
Q1-012	45662,78964	South	1016	Laptop	
Q1-013	45693,46014	East	1001	Printer	
Q1-014	45717,25416	South	1005	Printer	
Q1-015	45672,07266	South	1016	Monitor	
Q1-016	45742,87317	North	1003	Printer	
Q1-017	45688,21356	North	1008	Smartphone	
Q1-018	45718,93331	South	1009	Monitor	
Q1-019	45696,45154	North	1010	Headphones	
Q1-020	45729,8165	North	1004	Printer	

- o Ensure all dates are in UK format (DD/MM/YYYY).

The screenshot shows the Excel interface with the 'Sales_All' sheet. The data is organized in columns: OrderID, Date, Region, CustomerID, Product, Quantity, and UnitPrice. The 'Date' column contains dates in DD/MM/YYYY format. The 'Queries & Connections' pane on the right shows the data sources: Sales_Q1, Sales_Q2, Sales_Q3, and Append1 (90 rows loaded).

OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice
Q1-001	2025/02/20	South	1018	Tablet	1	730
Q1-002	2025/02/24	South	1004	Headphones	3	378
Q1-003	2025/03/20	West	1005	Monitor	1	585
Q1-004	2025/02/15	East	1018	Smartphone	8	726
Q1-005	2025/01/06	North	1012	Laptop	2	709
Q1-006	2025/02/28	East	1013	Laptop	8	1817
Q1-007	2025/03/19	South	1017	Tablet	9	1976
Q1-008	2025/02/19	South	1014	Tablet	5	1134
Q1-009	2025/01/14	South	1020	Printer	1	1220
Q1-010	2025/03/27	West	1010	Headphones	2	283
Q1-011	2025/01/27	West	1019	Laptop	9	417
Q1-012	2025/01/05	South	1016	Laptop	3	1692
Q1-013	2025/02/05	East	1001	Printer	4	1853
Q1-014	2025/03/01	South	1005	Printer	2	505
Q1-015	2025/01/15	South	1016	Monitor	3	1355
Q1-016	2025/03/26	North	1003	Printer	8	1220
Q1-017	2025/01/31	North	1008	Smartphone	3	605
Q1-018	2025/03/02	South	1009	Monitor	7	1149
Q1-019	2025/02/08	North	1010	Headphones	1	1621
Q1-020	2025/03/22	North	1004	Printer	3	1010
Q1-021	2025/02/25	South	1008	Headphones	7	241
Q1-022	2025/03/07	West	1005	Printer	7	1336
Q1-023	2025/01/05	West	1006	Laptop	3	1752
Q1-024	2025/02/27	East	1020	Headphones	8	911

- o Remove duplicate orders.



2. Customer Data Cleanup

- o Standardise names (use PROPER()).

F2	=CONCAT(B2;" ";PROPER(C2))							
	A	B	C	D	E	F	G	H
1	CustomerID	FirstName	LastName	Email	Region	Standardized		
2	1001	Naledi	Pillay	naledi.pillay@example.com	East	Naledi Pillay		
3	1002	Michael	van Wyk	michael.van wyk@example.com	East	Michael Van Wyk		
4	1003	Naledi	Botha	naledi.botha@example.com	East	Naledi Botha		
5	1004	Naledi	Botha	naledi.botha@example.com	North	Naledi Botha		
6	1005	Michael	Johnson	michael.johnson@example.com	North	Michael Johnson		
7	1006	Emily	Johnson	emily.johnson@example.com	South	Emily Johnson		
8	1007	Michael	Nguyen	michael.nguyen@example.com	West	Michael Nguyen		
9	1008	Sipho	Johnson	sipho.johnson@example.com	East	Sipho Johnson		
10	1009	Sarah	Nguyen	sarah.nguyen@example.com	North	Sarah Nguyen		
11	1010	John	Brown	john.brown@example.com	East	John Brown		
12	1011	John	Smith	john.smith@example.com	South	John Smith		

- o Extract **First Name**, **Last Name**, and **Email domain** (RIGHT, FIND).

A33		=LEFT(B2; FIND(" "; B2&" ") -1)					
	A	B	C	D	E		
1	CustomerID	FirstName	LastName	Email	Region	Standardized	
31	EXTRACTION						
32	First Name	Last Name	Email				
33	Naledi	Pillay	example.com				
34	Michael	van	example.com				
35	Naledi	Botha	example.com				
36	Naledi	Botha	example.com				
37	Michael	Johnson	example.com				
38	Emily	Johnson	example.com				
39	Michael	Nguyen	example.com				

- o Identify and flag duplicate customers with conditional formatting.

CustomerID	FirstName	LastName	Email	Region	Standardized
1003	Naledi	Botha	naledi.botha@example.com	East	Naledi Botha
1004	Naledi	Botha	naledi.botha@example.com	North	Naledi Botha
1005	Michael	Johnson	michael.johnson@example.com	North	Michael Johnson
1006	Emily	Johnson	emily.johnson@example.com	South	Emily Johnson
1007	Michael	Nguyen	michael.nguyen@example.com	West	Michael Nguyen
1008	Sipho	Johnson	sipho.johnson@example.com	East	Sipho Johnson
1009	Sarah	Nguyen	sarah.nguyen@example.com	North	Sarah Nguyen
1010	John	Brown	john.brown@example.com	East	John Brown
1011	John	Smith	john.smith@example.com	South	John Smith
1012	Naledi	van Wyk	naledi.van.wyk@example.com	East	Naledi Van Wyk
1013	Sipho	van Wyk	sipho.van.wyk@example.com	South	Sipho Van Wyk
1014	John	Mlambo	john.mlambo@example.com	East	John Mlambo
1015	John	Smith	john.smith@example.com	South	John Smith
1016	Sipho	Nguyen	sipho.nguyen@example.com	West	Sipho Nguyen
1017	Sipho	Smith	sipho.smith@example.com	North	Sipho Smith
1018	Michael	Mlambo	michael.mlambo@example.com	South	Michael Mlambo
1019	Naledi	Nguyen	naledi.nguyen@example.com	North	Naledi Nguyen
1020	David	van Wyk	david.van.wyk@example.com	West	David Van Wyk
1021	Thabo	Botha	thabo.botha@example.com	South	Thabo Botha
1022	Sipho	van Wyk	sipho.van.wyk@example.com	West	Sipho Van Wyk
1023	Lerato	Naidoo	lerato.naidoo@example.com	West	Lerato Naidoo
1024	Aisha	Brown	aisha.brown@example.com	East	Aisha Brown
1025	Lerato	van Wyk	lerato.van.wyk@example.com	East	Lerato Van Wyk
1015	John	Smith	john.smith@example.com	South	John Smith
1014	John	Mlambo	john.mlambo@example.com	East	John Mlambo
1018	Michael	Mlambo	michael.mlambo@example.com	South	Michael Mlambo

3. Product Data Cleanup

- o Merge Product_List with Sales_All using **VLOOKUP** or **XLOOKUP**.
- o Bring in Product Category, Unit Cost, and Discount %.

OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount
Q1-001	2025/02/20	South	1018	Tablet	1	730	Electronics	492	14%
Q1-002	2025/02/24	South	1004	Headphones	3	378	Accessories	323	10%
Q1-003	2025/03/20	West	1005	Monitor	1	585	Electronics	463	23%
Q1-004	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%
Q1-005	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%
Q1-006	2025/02/28	East	1013	Laptop	8	1817	Electronics	469	22%
Q1-007	2025/03/19	South	1017	Tablet	9	1976	Electronics	492	14%
Q1-008	2025/02/19	South	1014	Tablet	5	1134	Electronics	492	14%
Q1-009	2025/01/14	South	1020	Printer	1	1220	Office	136	10%

4. Finance & Operations Data

- o Link Finance_Data (Overheads & Salaries) to calculate **Net Profit**.

Month	Overheads	Salaries	OtherExpenses	NetProfit
2025/01/31	R 25 640,00	R 60 082,00	R 21 492,00	R 85 722,00
2025/02/28	R 43 320,00	R 70 533,00	R 27 724,00	R 113 853,00
2025/03/31	R 34 147,00	R 63 890,00	R 17 738,00	R 98 037,00
2025/04/30	R 44 423,00	R 90 674,00	R 21 152,00	R 135 097,00
2025/05/31	R 32 119,00	R 55 646,00	R 19 723,00	R 87 765,00
2025/06/30	R 20 079,00	R 99 954,00	R 15 769,00	R 120 033,00
2025/07/31	R 36 304,00	R 63 297,00	R 25 453,00	R 99 601,00
2025/08/31	R 36 734,00	R 93 760,00	R 16 230,00	R 130 494,00
2025/09/30	R 34 260,00	R 70 760,00	R 28 818,00	R 105 020,00

- o Link Operations_Data (Delivery Time, Returns) to measure **Operational Efficiency**.

OrderID	DeliveryDays	ReturnRate	Operational Efficiency
Q1-001	5	0,01	Needs Attention
Q1-002	2	0,03	Efficient
Q1-003	6	0,05	Needs Attention
Q1-004	4	0,07	Needs Attention
Q1-005	5	0,03	Needs Attention
Q1-006	5	0,06	Needs Attention
Q1-007	4	0,02	Needs Attention

Part 2: Advanced Calculations & Functions (2 hrs)

1. Create **calculated fields** in Sales_All:

- o Revenue = Quantity * Unit Price

K2											
=SUM(F2*G2)											
OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	
Q1-001	2025/02/20	South	1018	Tablet	1	730	Electronics	492	14%	730	
Q1-002	2025/02/24	South	1004	Headphones	3	378	Accessories	323	10%	1134	
Q1-003	2025/03/20	West	1005	Monitor	1	585	Electronics	463	23%	585	
Q1-004	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%	5808	
Q1-005	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%	1418	
Q1-006	2025/02/28	East	1013	Laptop	8	1817	Electronics	469	22%	14536	
Q1-007	2025/03/19	South	1017	Tablet	9	1976	Electronics	492	14%	17784	
Q1-008	2025/02/19	South	1014	Tablet	5	1134	Electronics	492	14%	5670	
Q1-009	2025/01/14	South	1020	Printer	1	1220	Office	136	10%	1220	
Q1-010	2025/03/27	West	1010	Headphones	2	283	Accessories	323	10%	566	
Q1-011	2025/01/27	West	1019	Laptop	9	417	Electronics	469	22%	3753	

- o Discounted Revenue = Revenue * (1 – Discount%)

L2	=SUM(K2*(1-J2))										
	D	E	F	G	H	I	J	K	L	M	
1	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue		
2	1018	Tablet	1	730	Electronics	492	14%	730	627,8		
3	1004	Headphones	3	378	Accessories	323	10%	1134	1020,6		
4	1005	Monitor	1	585	Electronics	463	23%	585	450,45		
5	1018	Smartphone	8	726	Electronics	97	8%	5808	5343,36		
6	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04		
7	1013	Laptop	8	1817	Electronics	469	22%	14536	11338,08		
8	1017	Tablet	9	1976	Electronics	492	14%	17784	15294,24		
9	1014	Tablet	5	1134	Electronics	492	14%	5670	4876,2		
10	1020	Printer	1	1220	Office	136	10%	1220	1098		
11	1010	Headphones	2	283	Accessories	323	10%	566	509,4		
12	1019	Laptop	9	417	Electronics	469	22%	3753	2927,34		

- o Profit = Revenue – (Quantity * Unit Cost) – Allocated Overheads

M2													
=SUM(K2- (@Quantity)*[@UnitCost]) - Finance!B3)													
A	B	C	D	E	F	G	H	I	J	K	L	M	N
OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	
Q1-001	2025/02/20	South	1018	Tablet	1	730	Electronics	492	14%	730	627,8	43 082,00	
Q1-002	2025/02/24	South	1004	Headphones	3	378	Accessories	323	10%	1134	1020,6	43 155,00	
Q1-003	2025/03/20	West	1005	Monitor	1	585	Electronics	463	23%	585	450,45	34 025,00	
Q1-004	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%	5808	5343,36	38 288,00	
Q1-005	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04	25 160,00	
Q1-006	2025/02/28	East	1013	Laptop	8	1817	Electronics	469	22%	14536	11338,08	9 295,00	
Q1-007	2025/03/19	South	1017	Tablet	9	1976	Electronics	492	14%	17784	15294,24	20 791,00	
Q1-008	2025/02/19	South	1014	Tablet	5	1134	Electronics	492	14%	5670	4876,2	40 110,00	
Q1-009	2025/01/14	South	1020	Printer	1	1220	Office	136	10%	1220	1098	24 556,00	
Q1-010	2025/03/27	West	1010	Headphones	2	283	Accessories	323	10%	566	509,4	80,00	
Q1-011	2025/01/27	West	1019	Laptop	9	417	Electronics	469	22%	3753	2927,34	468,00	
Q1-012	2025/01/05	South	1016	Laptop	3	1692	Electronics	469	22%	5076	3959,28	3 669,00	
Q1-013	2025/02/05	East	1001	Printer	4	1853	Office	136	10%	7412	6670,8	6 868,00	

2. Use Nested IFs / IFS to classify customers:

- o VIP: Revenue > 50,000
- o Regular: 20,000–50,000
- o Low Value: < 20,000

N2

=IF(K2>50000;"VIP";IF(K2>=20000;"Regular";"Low Value"))

OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	Nested IFS
Q1-026	2025/01/03	North	1009	Headphones	1	1841	Accessories	323	10%	1841	1656,9	24 122,00	Low Value
Q1-023	2025/01/05	West	1006	Laptop	3	1752	Electronics	469	22%	5256	4099,68	21 791,00	Low Value
Q1-012	2025/01/05	South	1016	Laptop	3	1692	Electronics	469	22%	5076	3959,28	21 971,00	Low Value
Q1-005	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04	25 160,00	Low Value
Q1-030	2025/01/09	East	1016	Monitor	5	1119	Electronics	463	23%	5595	4308,15	22 360,00	Low Value
Q1-009	2025/01/14	South	1020	Printer	1	1220	Office	136	10%	1220	1098	24 556,00	Low Value
Q1-015	2025/01/15	South	1016	Monitor	3	1355	Electronics	463	23%	4065	3130,05	22 964,00	Low Value
Q1-011	2025/01/27	West	1019	Laptop	9	417	Electronics	469	22%	3753	2927,34	26 108,00	Low Value
Q1-028	2025/01/28	West	1003	Monitor	6	1011	Electronics	463	23%	6066	4670,82	3 288,00	Low Value
Q1-017	2025/01/31	North	1008	Smartphone	3	605	Electronics	97	8%	1815	1669,8	24 116,00	Low Value
Q1-013	2025/02/05	East	1001	Printer	4	1853	Office	136	10%	7412	6670,8	36 452,00	Low Value
Q1-019	2025/02/08	North	1010	Headphones	1	1621	Accessories	323	10%	1621	1458,9	1 298,00	Low Value
Q1-004	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%	5808	5343,36	38 288,00	Low Value

3. Use INDEX-MATCH (not just VLOOKUP) to retrieve product details.

Q2

=INDEX(E2:E91;MATCH(H2;H2:H91;0))

OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	Nested IFS	Match & Index
Q1-026	2025/01/03	North	1009	Headphones	1	1841	Accessories	323	10%	1841	1656,9	24 122,00	Low Value	Headphones
Q1-023	2025/01/05	West	1006	Laptop	3	1752	Electronics	469	22%	5256	4099,68	21 791,00	Low Value	Laptop
Q1-012	2025/01/05	South	1016	Laptop	3	1692	Electronics	469	22%	5076	3959,28	21 971,00	Low Value	Laptop
Q1-005	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04	25 160,00	Low Value	Laptop
Q1-030	2025/01/09	East	1016	Monitor	5	1119	Electronics	463	23%	5595	4308,15	22 360,00	Low Value	Monitor
Q1-009	2025/01/14	South	1020	Printer	1	1220	Office	136	10%	1220	1098	24 556,00	Low Value	Printer
Q1-015	2025/01/15	South	1016	Monitor	3	1355	Electronics	463	23%	4065	3130,05	22 964,00	Low Value	Monitor
Q1-011	2025/01/27	West	1019	Laptop	9	417	Electronics	469	22%	3753	2927,34	26 108,00	Low Value	Laptop
Q1-028	2025/01/28	West	1003	Monitor	6	1011	Electronics	463	23%	6066	4670,82	3 288,00	Low Value	Monitor
Q1-017	2025/01/31	North	1008	Smartphone	3	605	Electronics	97	8%	1815	1669,8	24 116,00	Low Value	Smartphone
Q1-013	2025/02/05	East	1001	Printer	4	1853	Office	136	10%	7412	6670,8	36 452,00	Low Value	Printer
Q1-019	2025/02/08	North	1010	Headphones	1	1621	Accessories	323	10%	1621	1458,9	1 298,00	Low Value	Headphones
Q1-004	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%	5808	5343,36	38 288,00	Low Value	Smartphone
Q1-008	2025/02/19	South	1014	Tablet	5	1134	Electronics	492	14%	5670	4876,2	40 110,00	Low Value	Tablet

4. Create a Dynamic Named Range for Product Categories using OFFSET.

P2		=OFFSET(\$H\$2:0;0;COUNTA(\$H:\$H)-1;2)															
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
		Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	Nested IFS	Match & Index	Dynamic Range		
1	2025/01/03	North	1009	Headphones	1	1841	Accessories	323	10%	1841	1656,9	24 122,00	Low Value	Headphones	Accessories	323	
2	2025/01/05	West	1006	Laptop	3	1752	Electronics	469	22%	5256	4099,68	21 791,00	Low Value	Laptop	Electronics	469	
3	2025/01/05	South	1016	Laptop	3	1692	Electronics	469	22%	5076	3959,28	21 971,00	Low Value	Laptop	Electronics	469	
4	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04	25 160,00	Low Value	Laptop	Electronics	469	
5	2025/01/09	East	1016	Monitor	5	1119	Electronics	463	23%	5595	4308,15	22 360,00	Low Value	Monitor	Electronics	463	
6	2025/01/14	South	1020	Printer	1	1220	Office	136	10%	1220	1098	24 556,00	Low Value	Printer	Office	136	
7	2025/01/15	South	1016	Monitor	3	1355	Electronics	463	23%	4065	3130,05	22 964,00	Low Value	Monitor	Electronics	463	
8	2025/01/27	West	1019	Laptop	9	417	Electronics	469	22%	3753	2927,34	26 108,00	Low Value	Laptop	Electronics	469	
9	2025/01/28	West	1003	Monitor	6	1011	Electronics	463	23%	6066	4670,82	3 288,00	Low Value	Monitor	Electronics	463	
10	2025/01/31	North	1008	Smartphone	3	605	Electronics	97	8%	1815	1669,8	24 116,00	Low Value	Smartphone	Electronics	97	
11	2025/02/05	East	1001	Printer	4	1853	Office	136	10%	7412	6670,8	36 452,00	Low Value	Printer	Office	136	
12	2025/02/08	North	1010	Headphones	1	1621	Accessories	323	10%	1621	1458,9	1 298,00	Low Value	Headphones	Accessories	323	
13	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%	5808	5343,36	38 288,00	Low Value	Smartphone	Electronics	97	
14	2025/02/19	South	1014	Tablet	5	1134	Electronics	492	14%	5670	4876,2	40 110,00	Low Value	Tablet	Electronics	492	
15	2025/02/20	South	1018	Tablet	1	730	Electronics	492	14%	730	627,8	43 082,00	Low Value	Tablet	Electronics	492	

5. Apply **Array Formulas** / **Dynamic Arrays**:

- o Extract unique list of Regions (UNIQUE())

B96						
	A	B	C	D	E	F
94						
95		UNIQUE LIST				
96		North				
97		West				
98		South				
99		East				
100						

- o Count orders per Region (COUNTIF())

D96						
	A	B	C	D	E	F
94						
95		UNIQUE LIST		ORDERS		
96		North		22		
97		West		24		
98		South		23		
99		East		21		
100						

- o Average delivery time per Region (AVERAGEIF())

F96							
	A	B	C	D	E	F	G
94							
95		UNIQUE LIST		ORDERS		DELIVERY	
96		North		22		4,59	
97		West		24		5,08	
98		South		23		5,43	
99		East		21		4,38	

6. Use **TEXTJOIN** with criteria to display all products purchased by a single customer.

1. Create **Data Validation Drop-downs** for Region and Product Category.

North																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	Nested IFs	Match & Index	Dynamic R
1	Q1-026	2025/01/01	North	1009	Headphones	1	1841	Accessories	323	10%	1841	1656,90	1841	24.122,00	Low Value	Headphones
2	Q1-012	2025/01/01	West	1016	Laptop	3	1751	Electronics	229	469	5181	4909,50	1751	21.791,00	Low Value	Laptop
3	Q1-012	2025/01/01	West	1016	Laptop	3	1693	Electronics	229	5076	5076	3959,28	1693	21.973,00	Low Value	Laptop
4	Q1-005	2025/01/01	South	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04	709	25.160,00	Low Value	Laptop
5	Q1-030	2025/01/01	East	1016	Monitor	5	1119	Electronics	463	23%	5595	4308,15	1119	22.360,00	Low Value	Monitor

=XLOOKUP(E2; Products!A:A; Products!B:B; "Not Found")																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	Nested IFs	Match & Index	Dynamic Ranges
1	Q1-023	2025/01/05	West	1006	Laptop	3	1752	Electronics	469	22%	5218		4099.68	R	21 791.00	Low Value
2	Q1-012	2025/01/05	South	1012	Laptop	3	1692	Electronics	469	22%	5076		3959.28	R	21 971.00	Low Value
3	Q1-005	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%	1218		25 160.00	Low Value	Laptop	Electronics
4	Q1-030	2025/01/09	East	1016	Monitor	5	1119	Electronics	463	23%	5595		4308.15	R	22 360.00	Low Value
5	Q1-015	2025/01/15	South	1016	Monitor	3	1355	Electronics	463	23%	4005		3130.00	R	22 964.00	Low Value
6	Q1-011	2025/01/27	West	1019	Laptop	9	417	Electronics	469	22%	3763		2927.34	R	26 108.00	Low Value
7	Q1-028	2025/01/28	West	1003	Monitor	6	1011	Electronics	469	23%	6056		4670.82	R	3 288.00	Low Value
8	Q1-025	2025/01/31	North	1013	Smartphone	3	605	Electronics	87	8%	5115		24 116.00	R	1 669.00	Low Value
9	Q1-004	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%	5808		5343.36	R	38 288.00	Low Value
10	Q1-008	2025/02/19	South	1014	Tablet	5	1134	Electronics	492	14%	5670		4876.2	R	40 110.00	Low Value

[illegible]

4. Use **Remove Duplicates + Advanced Filter** to prepare a unique “VIP Customers” report.

OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	Nested Ifs	Match & Index	Dynamic Ra
Q1-026	2025/01/03	North	1009	Headphones	1	1841	Accessories	323	10%	1841	1656,9	24 122,00	Low Value	Headphones	Accessories
Q1-023	2025/01/05	West	1006	Laptop	3	1752	Electronics	469	22%	5256	4099,68	21 791,00	Low Value	Laptop	Electronics
Q1-012	2025/01/05	South	1016	Laptop	3	1692	Electronics	469	22%	5076	3959,28	21 971,00	Low Value	Laptop	Electronics
Q1-005	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04	25 160,00	Low Value	Laptop	Electronics
Q1-030	2025/01/09	East	1016	Monitor	5	1119	Electronics	463	23%	5595	4308,15	22 360,00	Low Value	Monitor	Electronics
Q1-009	2025/01/14	South	1020	Printer	1	1220	Office	136	10%	1220	1098	24 556,00	Low Value	Printer	Office
Q1-015	2025/01/15	South	1016	Monitor	3	1355	Electronics	463	23%	4065	3130,05	22 964,00	Low Value	Monitor	Electronics
Q1-011	2025/01/27	West	1013	Laptop	9	417	Electronics	469	22%	3753	2927,34	26 108,00	Low Value	Laptop	Electronics
Q1-028	2025/01/28	West	1003	Monitor	6	1011	Electronics	463	23%	6066	4670,82	3 288,00	Low Value	Monitor	Electronics
Q1-017	2025/01/31	North	1008	Smartphone	3	605	Electronics	97	8%	1815	1669,8	24 116,00	Low Value	Smartphone	Electronics
Q1-013	2025/02/05	East	1001	Printer	4	1853	Office	136	10%	7412	6670,8	36 452,00	Low Value	Printer	Office
Q1-019	2025/02/08	North	1010	Headphones	1	1621	Accessories	323	10%	1621	1458,9	1 298,00	Low Value	Headphones	Accessories
Q1-004	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%	5808	5343,36	38 288,00	Low Value	Smartphone	Electronics
Q1-008	2025/02/19	South	1014	Tablet	5	1134	Electronics	492	14%	5670	4876,2	40 110,00	Low Value	Tablet	Electronics
Q1-001	2025/02/20	South	1018	Tablet	1	730	Electronics	492	14%	730	627,8	40 110,00	Low Value	Tablet	Electronics
Q1-025	2025/02/22	South	1007	Monitor	8	1770	Electronics	463	23%	14160	10903,2	10 456,00	Low Value	Monitor	Electronics

Part 4: Conditional Formatting (30 mins)

1. Highlight Top 10 Customers by Revenue.

OrderID	Date	Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	Nested Ifs	Match & Index	Dynamic Ra
Q1-026	2025/01/03	North	1009	Headphones	1	1841	Accessories	323	10%	1841	1656,9	24 122,00	Low Value	Headphones	Accessories
Q1-023	2025/01/05	West	1006	Laptop	3	1752	Electronics	469	22%	5256	4099,68	21 791,00	Low Value	Laptop	Electronics
Q1-012	2025/01/05	South	1016	Laptop	3	1692	Electronics	469	22%	5076	3959,28	21 971,00	Low Value	Laptop	Electronics
Q1-005	2025/01/06	North	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04	25 160,00	Low Value	Laptop	Electronics
Q1-030	2025/01/09	East	1016	Monitor	5	1119	Electronics	463	23%	5595	4308,15	22 360,00	Low Value	Monitor	Electronics
Q1-009	2025/01/14	South	1020	Printer	1	1220	Office	136	10%	1220	1098	24 556,00	Low Value	Printer	Office
Q1-015	2025/01/15	South	1016	Monitor	3	1355	Electronics	463	23%	4065	3130,05	22 964,00	Low Value	Monitor	Electronics
Q1-011	2025/01/27	West	1013	Laptop	9	417	Electronics	469	22%	3753	2927,34	26 108,00	Low Value	Laptop	Electronics
Q1-028	2025/01/28	West	1003	Monitor	6	1011	Electronics	463	23%	6066	4670,82	3 288,00	Low Value	Monitor	Electronics
Q1-017	2025/01/31	North	1008	Smartphone	3	605	Electronics	97	8%	1815	1669,8	24 116,00	Low Value	Smartphone	Electronics
Q1-013	2025/02/05	East	1001	Printer	4	1853	Office	136	10%	7412	6670,8	36 452,00	Low Value	Printer	Office
Q1-019	2025/02/08	North	1010	Headphones	1	1621	Accessories	323	10%	1621	1458,9	1 298,00	Low Value	Headphones	Accessories
Q1-004	2025/02/15	East	1018	Smartphone	8	726	Electronics	97	8%	5808	5343,36	38 288,00	Low Value	Smartphone	Electronics
Q1-008	2025/02/19	South	1014	Tablet	5	1134	Electronics	492	14%	5670	4876,2	40 110,00	Low Value	Tablet	Electronics
Q1-001	2025/02/20	South	1018	Tablet	1	730	Electronics	492	14%	730	627,8	40 110,00	Low Value	Tablet	Electronics
Q1-025	2025/02/22	South	1007	Monitor	8	1770	Electronics	463	23%	14160	10903,2	10 456,00	Low Value	Monitor	Electronics
Q1-002	2025/02/24	South	1004	Headphones	3	378	Accessories	323	10%	1134	1020,6	43 155,00	Low Value	Headphones	Accessories
Q1-021	2025/02/25	South	1008	Headphones	7	241	Accessories	323	10%	1687	1518,3	574,00	Low Value	Headphones	Accessories
Q1-024	2025/02/27	East	1020	Headphones	8	911	Accessories	323	10%	7288	6559,2	4 704,00	Low Value	Headphones	Accessories
Q1-006	2025/02/28	East	1013	Laptop	8	1817	Electronics	469	22%	14536	11338,08	9 295,00	Low Value	Laptop	Electronics

2. Use a Formula Rule:

o Highlight customers whose **Revenue > 25,000 AND Returns > 5%**.

3. Apply **Icon Sets** for Delivery Performance (Green = On Time, Red = Late).

Region	CustomerID	Product	Quantity	UnitPrice	Category	UnitCost	Discount	Revenue	Discounted Revenue	Profit	Nested Ifs	Match & Index	Dynamic Range	DeliveryDays
03 North	1009	Headphones	1	1841	Accessories	323	10%	1841	1656,9	24 122,00	Low Value	Headphones	Accessories	323
05 West	1006	Laptop	3	1752	Electronics	469	22%	5256	4099,68	21 791,00	Low Value	Laptop	Electronics	469
05 South	1016	Laptop	3	1692	Electronics	469	22%	5076	3959,28	21 971,00	Low Value	Laptop	Electronics	469
06 North	1012	Laptop	2	709	Electronics	469	22%	1418	1106,04	25 160,00	Low Value	Laptop	Electronics	469
09 East	1016	Monitor	5	1119	Electronics	463	23%	5595	4308,15	22 360,00	Low Value	Monitor	Electronics	463
14 South	1020	Printer	1	1220	Office	136	10%	1220	1098	24 556,00	Low Value	Printer	Office	136
15 South	1016	Monitor	3	1355	Electronics	463	23%	4065	3130,05	22 964,00	Low Value	Monitor	Electronics	463
17 West	1013	Laptop	9	417	Electronics	469	22%	3753	2927,34	26 108,00	Low Value	Laptop	Electronics	469
18 West	1003	Monitor	6	1011	Electronics	463	23%	6066	4670,82	3 288,00	Low Value	Monitor	Electronics	463
19 North	1008	Smartphone	3	605	Electronics	97	8%	1815	1669,8	24 116,00	Low Value	Smartphone	Electronics	97
19 East	1001	Printer	4	1853	Office	136	10%	7412	6670,8	36 452,00	Low Value	Printer	Office	136
19 North	1010	Headphones	1	1621	Accessories	323	10%	1621	1458,9	1 298,00	Low Value	Headphones	Accessories	323
15 East	1018	Smartphone	8	726	Electronics	97	8%	5808	5343,36	38 288,00	Low Value	Smartphone	Electronics	97
15 South	1014	Tablet	5	1134	Electronics	492	14%	5670	4876,2	40 110,00	Low Value	Tablet	Electronics	492

4. Apply a **3-Colour Scale** to Expenses (Low–Medium–High).

D1 OtherExpenses					
	A	B	C	D	E
1	Month	Overheads	Salaries	OtherExpenses	NetProfit
2	2025/01/31	R 25 640,00	R 60 082,00	R 21 492,00	R 85 722,00
3	2025/02/28	R 43 320,00	R 70 533,00	R 27 724,00	R 113 853,00
4	2025/03/31	R 34 147,00	R 63 890,00	R 17 738,00	R 98 037,00
5	2025/04/30	R 44 423,00	R 90 674,00	R 21 152,00	R 135 097,00
6	2025/05/31	R 32 119,00	R 55 646,00	R 19 723,00	R 87 765,00
7	2025/06/30	R 20 079,00	R 99 954,00	R 15 769,00	R 120 033,00
8	2025/07/31	R 36 304,00	R 63 297,00	R 25 453,00	R 99 601,00
9	2025/08/31	R 36 734,00	R 93 760,00	R 16 230,00	R 130 494,00
10	2025/09/30	R 34 260,00	R 70 760,00	R 28 818,00	R 105 020,00

Part 5: Advanced PivotTables & Analysis (1.5 hrs)

1. Build PivotTables for:

o Revenue & Profit by Region → with slicers

Sum of Revenue					Revenue		Profit		Region	
Row Labels	East	North	South	West	Grand Total					
-R 43 155,00			1134		1134	218	-R43 155,00		East	
-R 43 082,00			730		730	233	-R43 082,00		North	
-R 40 110,00			5670		5670	357	-R40 110,00		South	
-R 38 288,00	5808				5808	398	-R38 288,00		West	
-R 36 452,00	7412				7412	432	-R36 452,00			
-R 34 025,00				585	585	444	-R34 025,00			
-R 26 108,00				3753	3753	566	-R26 108,00			
-R 25 160,00		1418			1418	585	-R25 160,00			
-R 24 556,00			1220		1220					
-R 24 122,00		1841			1841					
-R 24 116,00		1815			1815					
-R 22 964,00			4065		4065					

o Product Category vs Net Profit → with conditional formatting

Row Labels	Sum of Profit
Accessories	-R 1 694,00
Electronics	-R213 951,00
Office	-R 25 089,00
Grand Total	-R240 734,00

o Customer Classification (VIP, Regular, Low Value)

- o Delivery Performance → Average Delivery Days by Region

Row Labels	Sum of DeliveryDays
East	92
North	101
South	125
West	122
Grand Total	440

2. Insert **Calculated Fields** inside PivotTables for:

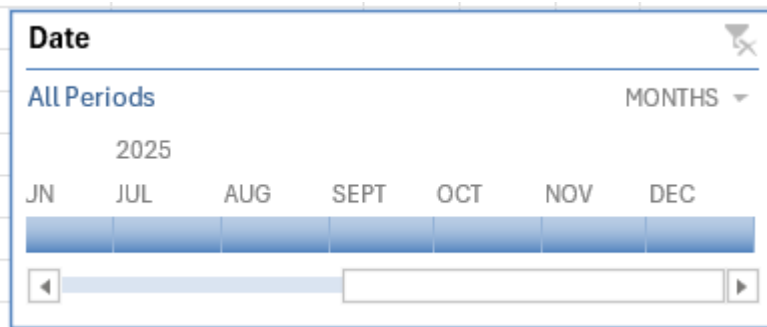
- o Profit Margin %

- o Return Rate %

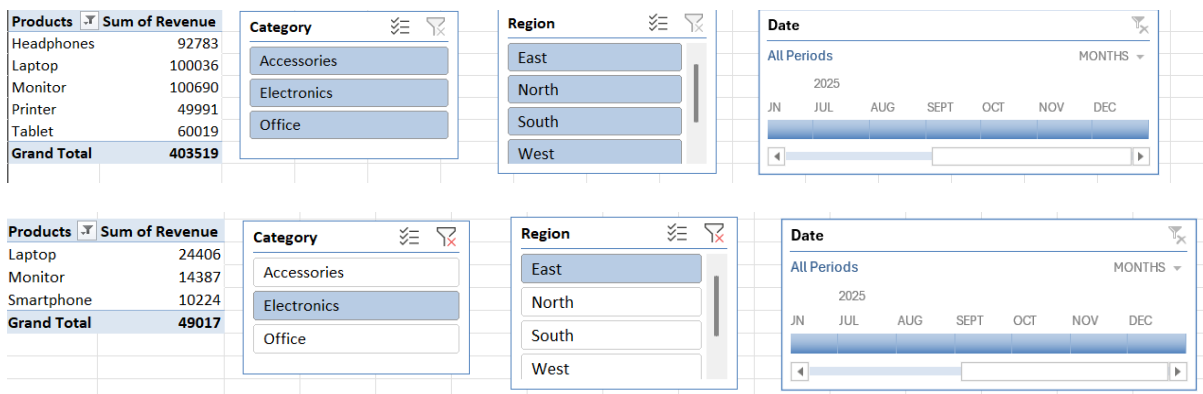
Row Labels	Sum of Revenue	Sum of Profit	Sum of Profit Margin %
Headphones	92783 -R	1 694,00	-1,83%
Laptop	100036 -R	70 910,00	-70,88%
Monitor	100690 -R	25 944,00	-25,77%
Printer	49991 -R	25 089,00	-50,19%
Smartphone	37743 -R	36 649,00	-97,10%
Tablet	60019 -R	80 448,00	-134,04%
Grand Total	441262 -R	240 734,00	-54,56%

3. Add **Slicers & Timelines** for Month, Region, and Category.

Count of Category	Column Labels				
Row Labels	East	North	South	West	Grand Total
Jan	1	3	3	3	10
Feb	4	1	5		10
Mar	1	2	3	4	10
Apr	5	4	1	4	14
May	2	3	2	2	9
Jun	2	1	1	3	7
Jul	2	3	3	4	12
Aug	2	5	2	2	11
Sept	2		3	2	7
Grand Total	21	22	23	24	90

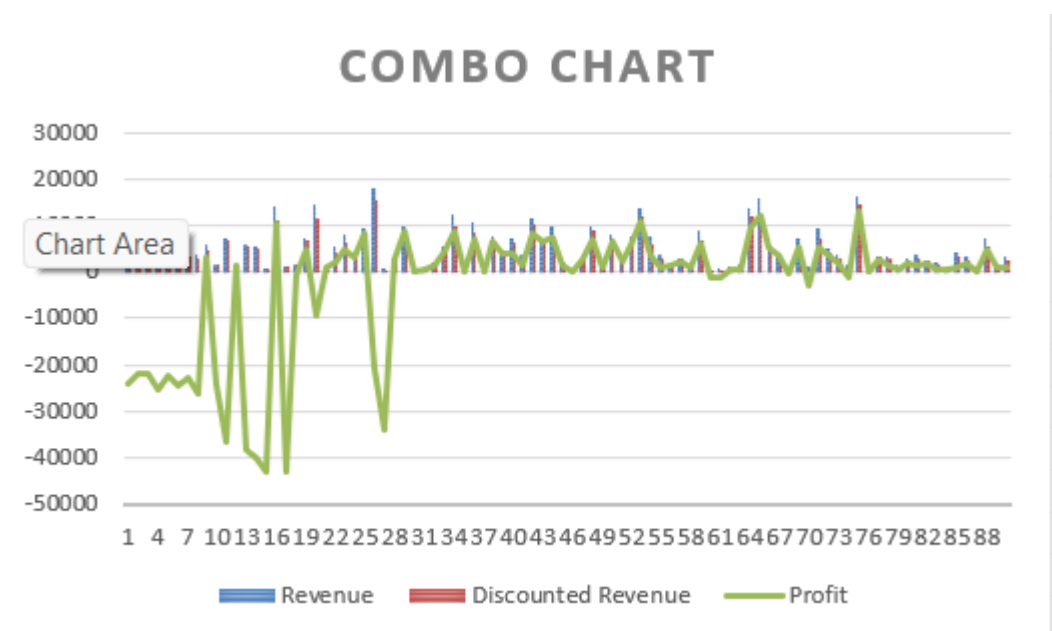


4. Show **Top 5 Products by Revenue** with a dynamic filter.

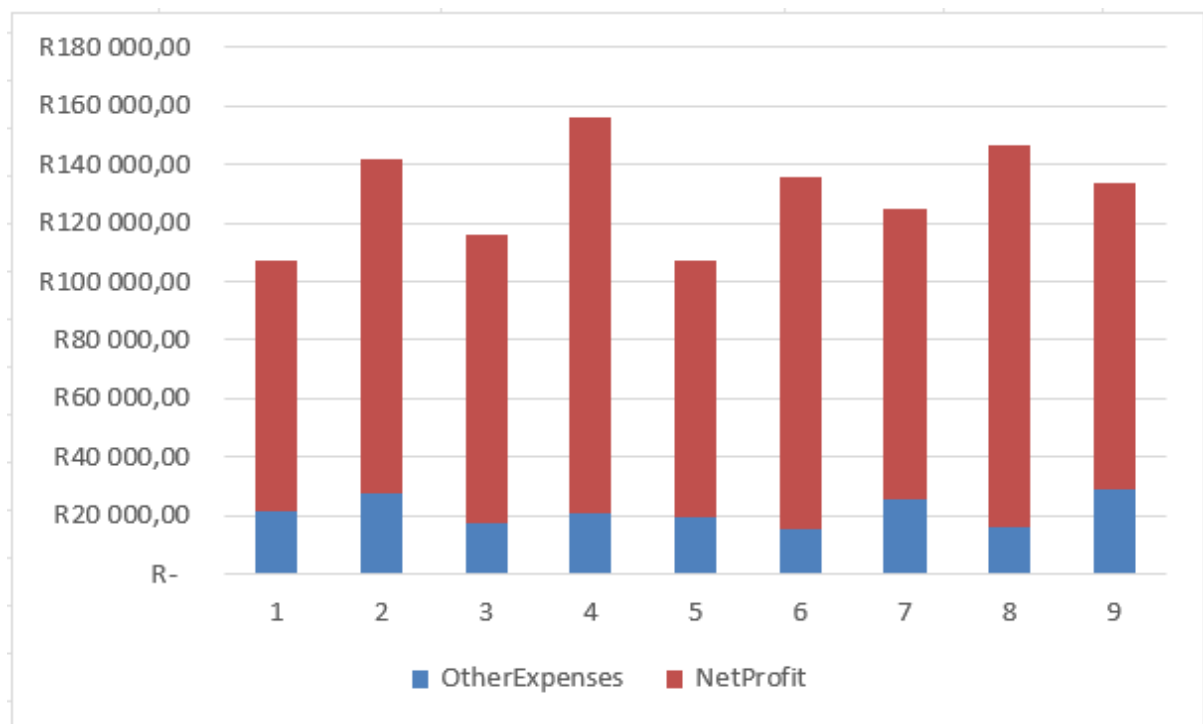


Part 6: Charts & Dashboard (2 hrs)

- Create the following charts:
 - Combo Chart:** Revenue (Columns) + Profit Margin % (Line)



- o **Map Chart (if available):** Sales by Region
- o **Stacked Bar Chart:** Expenses vs Net Profit



- o **Pareto Chart:** 80/20 rule for Products (Top 20% of products contributing to 80% of revenue)

2. Build a **Dashboard Sheet** with:

- o KPIs (Total Revenue, Profit, Top Customer, On-Time Delivery %)

A2 ✕ ✓ <i>fx</i> =SUM(Append1[Discounted Revenue])					
	A	B	C	D	E
1	TotalRevenue	TotalProfit	TopCustomer		
2	R 370 395,88	-R 240 734,00	1017		

- o Interactive slicers & timelines

Count of Product	Column Labels				
Row Labels	East	North	South	West	Grand Total
Accessories	6	4	4	3	17
Electronics	11	13	15	19	58
Office	4	5	4	2	15
Grand Total	21	22	23	24	90

Category

- Accessories
- Electronics
- Office

Product

- Headphones
- Laptop
- Monitor
- Printer
- Smartphone
- Tablet

Region

- East
- North
- South
- West

Date

All Periods
MONTHS

2025

JN JUL AUG SEPT OCT NOV DEC

o Clean, professional formatting

3. Add **Dynamic Headings** (e.g., “Dashboard for [Selected Region]”) using cell references linked to slicers.

Slicer Settings

Source Name: Region
Name to use in formulas: Slicer_Region1
Name: Slicer_Region1

Header
☒ Display header
Caption: Region

Item Sorting and Filtering

☒ Ascending (A to Z)
☐ Descending (Z to A)
☒ Use Custom Lists when sorting

☐ Hide items with no data
☒ Visually indicate items with no data
☒ Show items with no data last
☒ Show items deleted from the data source

OK
Cancel